

Abhijeet Minz

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Research Summary

PhD candidate in applied mathematics specialising in geophysical fluid dynamics. Expertise in Lagrangian averaging within the Generalised Lagrangian Mean (GLM) framework, with strong background in geometric formulation of fluid models, asymptotic analysis, wave-mean flow interaction, and numerics using Fourier pseudo-spectral solvers in python.

Research interests: Geophysical fluid dynamics, climate and ocean modelling, computational mathematics.

Education

PhD - MAC-MIGS CDT Programme Sept 2022 – Sept 2026 (expected)
Mathematical Modelling, Analysis and Computation (PhD with Integrated Study)

Thesis title: Lagrangian averaging with exponential filters

Supervisors: Jacques Vanneste (primary) (UoE) & Lois Baker (UoE)

- **Relevant Coursework:**

- Geometric Fluid Dynamics (variational principles, conservation laws)
- Numerical Methods for Partial Differential Equations
- Finite Element Methods (weak formulations, Galerkin methods)
- Asymptotic Methods and Perturbation Theory
- Differential Topology
- Developed models for Lagrangian averaging in the GLM framework
- Implemented FFT-based pseudo-spectral solvers for rotating shallow water equations
- Computed Lagrangian mean quantities and pseudomomentum fields in nonlinear regimes
- Industrial project: Spatio-temporal modelling of aphid-parasitoid dynamics for James Hutton institute
- Taster project in training in data assimilation and inverse problems

MSc in Mathematics (Awarded distinction) 2018 – 2020
Bangalore University, India

- Dissertation: Basset Memory Integral. Analyzed the Maxey–Riley (MR) intro-differential equation which governs the motion of small, finite-sized spherical particles in fluid flow, with a specific focus on the Basset history term.

BSc in Physics, Maths and Computer Science (PMC) (Awarded distinction) 2015 – 2018
Bangalore University, India

- Projects: A study of Maxwell’s equations; and developed a user-interface DBMS using visual studio, visual basic as the frontend and MYSQL as the backend.

Publications

- **Minz, A.**, Baker, L. E., Kafiabad, H. A., & Vanneste, J. (2025). Efficient Lagrangian averaging with exponential filters. *Physical Review Fluids*, 10, 074902 (2025)
DOI: 10.1103/4d4f-38wx

Manuscript in preparation

- **Minz, A.**, Baker, L. E., & Vanneste, J. Volume-preserving Lagrangian averaging using polar factorisation (to be submitted to the *Physical Review Fluids* estimated in May 2026)

Software and Data Products

- **Minz, A.**, Baker, L., Kafiabad, H., & Vanneste, J. (2025). Code and data for "Efficient Lagrangian averaging with exponential filters" (1.0.1). DOI: 10.5281/zenodo.15213015

Technical Skills

Programming: Python (NumPy, SciPy, Matplotlib), MATLAB (basic), Mathematica

Languages: C++, C, Java, SQL

Methods: FFT-based pseudo-spectral methods, numerical PDE solvers, FEM, FDM

Tools: LaTeX, Git, Linux

Conferences and Talks

- Ocean Science Meeting, Glasgow Feb 2025
Poster: *Volume-preserving Lagrangian averaging using polar factorisation*
- ICTW2025, Durham University July 2025
Talk: *Lagrangian averaging*
- North-East GFD Meeting, Edinburgh June 2025
Talk: *Exponential Lagrangian averaging*
- Scottish Fluid Mechanics Meeting, Glasgow May 2025
Talk: *Exponential Lagrangian averaging*
- North-East GFD Meeting, Edinburgh June 2024
Talk: *Lagrangian averaging of tensor fields*
- MAC-MIGS / Utrecht / CWI Workshop May 2024
Talk: *Lagrangian averaging of scalar fields*

Teaching Experience

Tutor, University of Edinburgh

- **Tutor, Introduction to Mathematics at University**, UoE Sept 2025 – Nov 2025
– Taught and assessed mathematical reasoning, proof techniques, and foundational concepts
- **Tutor, Numerical ODEs**, UoE Jan 2025 – March 2025
– Focused on implementing numerical solvers for differential equations in Python
- **Tutor, Introduction to Linear Algebra**, UoE Sept 2023 – Nov 2023
– Led weekly tutorials (10–15 students), facilitating problem-solving and discussion
- **Tutor, Honours Differential Equations**, UoE Sept 2023 – Nov 2023
– Supported development of analytical techniques and exam preparation
- **Tutor, Mathematics** (Outreach program), Mount Fort School, India 2021

Additional Information

Competitive athlete (sprinting and football, 2013–2018)